

BIODIVERSITY

The word biodiversity is a combination of two words: “biological and diversity” and refers to the variety of life on the Earth. **Biodiversity** is the degree of variation of life forms within a given species, ecosystem, biome, or an entire planet. Biodiversity is a measure of the health of ecosystems.

The term biological diversity was used first by wildlife scientist and conservationist Raymond F. Dasmann in the 1968.

The term's contracted form biodiversity may have been coined by W.G. Rosen in 1985

TYPES OF BIODIVERSITY :

The following are different types of biodiversity

1. Genetic Diversity: A species with different genetic characteristics is known as a sub-species or "genera".

Genetic diversity is a measure of the variety of versions of same gene within individual species. Within individual species, there are varieties, that are slightly different from one other. These differences are due to differences in the combination of genes.

Genes are the basic units of hereditary information transmitted from one generation to the other.

Ex: (i) Rice varieties - All rice varieties belong to the species "oryzasativa". However, there are thousands of rice varieties that show variation at the genetic level in the form of different size, shape, colour and nutrient content. (ii) Teak wood varieties: The various teak wood varieties available are - Indian teak, Burma teak, Malaysian teak etc

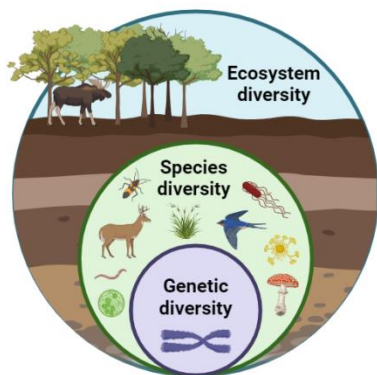
2. Species Diversity: A discrete group of organisms of the same kind is known as species.

Species diversity is the diversity between different species.

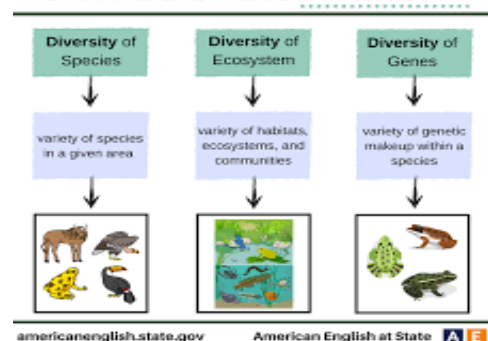
Species diversity refers to the variety of different types of species found in a particular area. It is the biodiversity at the most basic level. It includes all the species ranging from plants to different microorganism.

No two individuals of the same species are exactly similar. For example, humans show a lot of diversity among themselves .

3. Ecosystem Diversity: Ecological diversity or ecosystem diversity is the variety of biological communities, such as forests, deserts, grasslands and streams that interact with one another and with their physical and chemical (nonliving) environments. It relates to the different forms of life which are present in any one particular area or site, in more precise terms, it concerns the different species of a particular genus which are present in an ecological community.



3 TYPES OF BIODIVERSITY



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Fig1: Types of Biodiversity

VALUES OF BIODIVERSITY :

The value of biodiversity (in terms of its commercial utility, ecological services, social and aesthetic values) is enormous. There are several ways that biodiversity and its various forms are Valuable to humans. The biodiversity value may be classified as follows: The value of biodiversity is classified into:

1. Direct Values

2. Indirect Values

Direct values

1. Consumptive value
2. Productive value

1. Consumptive value: Biodiversity is an essential requirement for the maintenance of global food supply. The main sources of human food include animals, fish and plant produces. A large number of plants are consumed by human beings as food. A few animal species are consumed by people which come from cattle, pigs, sheep, goats, buffaloes, chickens, ducks, geese and turkey species.

Fish: Many fresh water fish can be grown in ponds. Israel and China already get about half of their fish from aqua culture.

Drugs & medicines: About 75% of the world's population depends upon plants or plant extracts for medicines. The drug Penicillin used as an antibiotic is derived from a fungus called **Penicillium**. Likewise, Quinine which is used to cure malaria is obtained from the bark of cinchona tree. .

Fuel: The fossil fuels like coal, petroleum products and natural gas are the products of biodiversity.

2. Productive value: Some of the organisms are commercially useful where the product is marketed and sold. The animal products like tusks of elephants; musk from deer; silk from silkworm; wool from sheep or goats; fur of many animals etc all of which are traded in the market.

Timber, fish, honey, construction materials, mushrooms, fruits, medicinal plants and game meat sold in a market have productive use value.

Indirect values

3. Social value: These are the values associated with the social life, religion and spiritual aspects of the people. Many of the plants are considered to be sacred in our country like Tulasi, Mango leaves, Banana leaves . The leaves, fruits, flowers of some of the plants are used in worship. Many animals like cow, snake, bull, peacock also have significant place in spiritual and thus hold special importance. Thus, biodiversity has distinct social value, attached with different societies.

4. Ethical value: The ethical value means that human beings may or may not use a certain species but knowing the very fact that this species exists in nature gives pleasure.

For eg: A peculiar species of Pigeon, grey / white bird with short legs is no more on this earth. Similarly, Dodo species is also no more. Human beings are not deriving anything direct from Kangaroo, giraffe but strongly feel that these species should exist in nature.

5. Aesthetic value: Every one of us would like to visit vast stretches of lands to enjoy the visible life. People from farther areas, spend a lot of time and money to visit wild life areas where they can enjoy the aesthetic value of biodiversity and this type of tourism is known as eco tourism.

Eco-tourism is estimated to generate 12 billion dollars of revenue annually that roughly gives the aesthetic value of biodiversity.

A study of the impact of environment on the psyche was undertaken by Kaplan and Kaplan (1989) in which they found that being near nature relieved working stresses while people who worked in closed environment or human made structures experienced much more job stresses and illnesses.

6.Optional value:

This refers to the potential of biodiversity that is currently known and needs to be explored.

This refers to the idea that there may be several existing species that may prove to be important in future and their usefulness needs to be studied with reference to a specific problem currently plaguing the society.

Ex: 1. There is a possibility that we may have some potential cure for diseases like cancer and AIDS existing within the depths of marine ecosystem or a Tropical rain forests.

Medicinal plants and herbs play a very important role in the economic growth of our country.

BIODIVERSITY AT GLOBAL, NATIONAL AND LOCAL LEVEL

The enormous diversity of life forms in the biosphere has evolved essentially through the process of trial and error during course of organic evolution. The changes in character of living organism which confer some advantage to the species are retained.

The changes in climatic conditions are reflected in the distribution of living organism and the pattern of biodiversity on our planet. The number of species present per unit area decreases as we move from mild tropics to the tundra's.

The Indian region (8° to 30° N and 60° to 97.5°) with total area of 329 million hectares is very rich in biodiversity. It is estimated that about 4500 species of plants occur in this country. The position of Indian sub-continent at the confluence of there biogeography reels is also an important contributing factor and explain the preserve of African, European, Sind, Japanese and Indo-Malayan elements in the flora and fauna in India. It is the sum total of such remarkable diversity that has made India a "gene bank" for a number of food crops, forest trees, medical and aromatic plants and domesticated animal.

Forests are important bioreserves; most of the 1700 million hectares of tropical forests are located in poor countries. The forests surrounding Reo de Aneroid are part of vegetation which is rich in species of plants and animals that are endemic. There are about 53.5% of trees species found only in these forests and studies of birds, reptiles, primates and butter flies have revealed equally high or higher endemics.

INDIA AS A MEGA DIVERSITY NATION :

India contains a great wealth of biodiversity in the forests, wet lands and marine areas. Hence biodiversity can be observed at all levels ie locally, nationally and globally . India, as a subcontinent representing a major part of South Asia is rich in flora and fauna and hence it is one of the world's "MEGADIVERSITY NATIONS" .

It is estimated that over 75000 species of animals and over 45000 species of plants are found in India.

Biogeographic regions of India: According to **wild life Institute of India**, the country has 10 distinct biogeographic zones or regions. They are:

1. Trans – Himalayan Zone
2. Himalayan Zone
3. Desert Zone
4. Semi – arid Zone
5. Western Ghats
6. Deccan Zone
7. Gangetic plain Zone
8. NE Indian Zone
9. Coastal Zone
10. Islands around the country

HOT SPOTS OF BIODIVERSITY:

A biodiversity hotspot is a region with significant biodiversity that is under threat. There are 36 recognized biodiversity hotspots around the world, covering about 2.5% of the Earth's land surface

Species which are restricted only to particular areas are known as endemic.

India shows a good number of endemic species. About 62% of amphibians and 50% of lizards are endemic to India. Western Ghats are the site of maximum endemism. The term “Hot spots” was introduced by **Myers** (1988).

terrestrial biodiversity. According to Myers an area is designated as a hotspot when it contains at least 0.5% of the plant species as endemics.

Following the criteria for an area to be declared as a Biodiversity Hotspot, there are major four biodiversity hotspots in India:

1. The Himalayas
2. Indo-Burma Region
3. The Western Ghats
4. Sundaland

The Himalayas

Considered the highest in the world, the Himalayas (overall) comprise North-East India, Bhutan, Central and Eastern parts of Nepal. This region (NE Himalayas) holds a record of having 163 endangered species which includes the Wild Asian Water Buffalo, One-horned Rhino; and as many as 10,000 plant species, of which 3160 are endemic. This mountain range covers nearly 750,000 km².

Indo – Burma Region

The Indo-Burma Region is stretched over a distance of 2,373,000 km². In the last 12 years, 6 large mammal species have been discovered in this region: the Large-antlered Muntjac, the Annamite Muntjac, the Grey-shanked Douc, the Annamite Striped Rabbit, the Leaf Deer, and the Saola.

This hotspot is also known for the endemic freshwater turtle species, most of which are threatened with extinction, due to over-harvesting and extensive habitat loss. There are also

1,300 different bird species, including the threatened White-eared Night-heron, the Grey-crowned Crocias, and the Orange-necked Partridge.

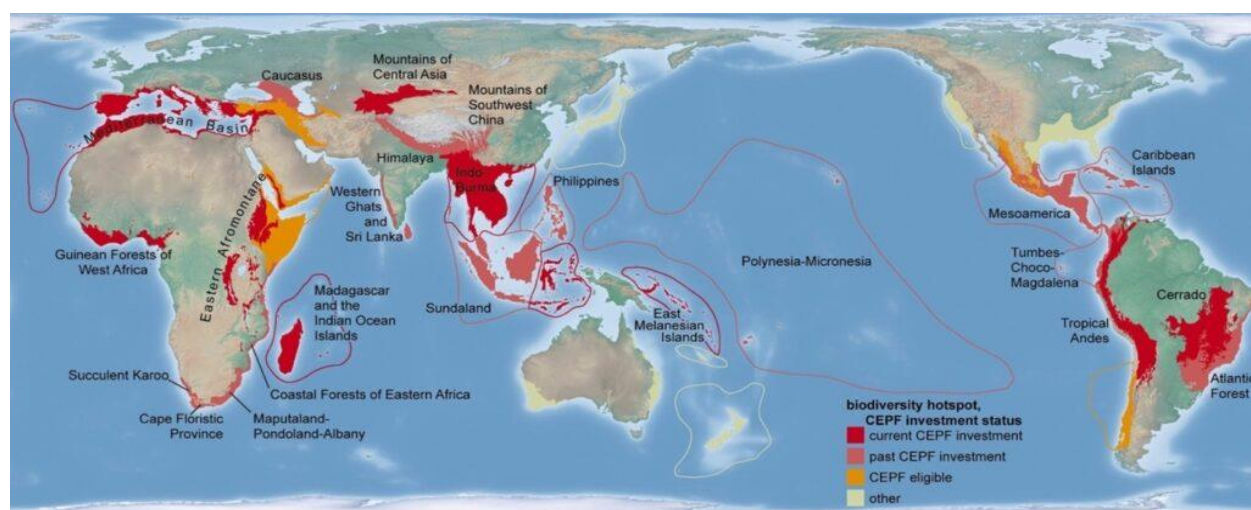
The Western Ghats

The Western Ghats are present along the western edge of peninsular India and cover most of the deciduous forests and rain forests. As per UNESCO, it is home to at least 325 globally threatened flora, fauna, bird, amphibian, reptile and fish species. Originally, the vegetation in this region was spread over 190,000 km² but has been now reduced to 43,000 km². The region is also known for the globally threatened flora and fauna represented by 229 plant species, 31 mammal species, 15 bird species, 43 amphibian species, 5 reptile species and 1 fish species. UNESCO mentions that “Of the total 325 globally threatened species in the Western Ghats, 129 are classified as Vulnerable, 145 as Endangered and 51 as Critically Endangered.”

Knowing in detail about the Western Ghats will be helpful for the aspirants for the Geography preparation.

Sundaland

The Sundaland hotspot lies in Southeast Asia and covers Singapore, Thailand, Indonesia, Brunei, and Malaysia. The Sundaland biodiversity hotspot includes the Andaman and Nicobar Islands, specifically representing India within this region. In the year 2013, the Sundaland was declared as a World Biosphere Reserve by the United Nations. This region is famous for its rich terrestrial and marine ecosystem. Sundaland is one of the biologically richest hotspots in the world which comprises 25,000 species of vascular plants, of which 15,000 are found only in this region.



hotspots of biodiversity in world

THREATS TO BIODIVERSITY :

Extinction or elimination of a species is a natural process of evolution. In the geologic period the earth has experienced mass extinctions. During evolution, species have died out and have been replaced by others. However, the rate of loss of species in geologic past has been a slow process, keeping in view the vast span of time going back to 444 million years. The process of extinction has become particularly fast in the recent years of civilization. Edward O. Wilson prefers the acronym HIPPO, standing for habitat destruction, invasive species, pollution, human overpopulation, and over-harvesting

Following are the major causes and issues related to threats to biodiversity

1. Habitat destruction: Habitat destruction has played a key role in extinctions, especially related to tropical forest destruction. Factors contributing to habitat loss are: overpopulation, deforestation, pollution (air pollution, water pollution, soil contamination) and global warming or climate change.

Habitat size and numbers of species are systematically related. Physically larger species and those living at lower latitudes or in forests or oceans are more sensitive to reduction in habitat area. Conversion to "trivial" standardized ecosystems (e.g., monoculture following deforestation) effectively destroys habitat for the more diverse species that preceded the conversion. In some countries lack of property rights or lax law/regulatory enforcement necessarily leads to biodiversity loss (degradation costs having to be supported by the community)

2. Poaching: Illegal trade of wildlife products by killing prohibited endangered animals i.e. poaching is another threat to wildlife. Despite international ban on trade in products from endangered species, smuggling of wildlife items like furs, hides, horns, tusks, live specimens and herbal products worth millions of dollars per year continues, the developing nations in Asia, Latin America and Africa are the richest source of biodiversity and have enormous wealth of wildlife. The rich countries in Europe and North America and some affluent countries in Asia like Japan, Taiwan and Hong Kong are the major importers of the wildlife products or wildlife itself. The trading of such wild life products is highly profit making for the poachers who just hunt these prohibited wild lives and smuggle it to other countries mediated through mafia. The worst part is that for every live animal that actually gets into the market about 50 additional animals are caught and killed.

If you are fond of rare plants, fish or birds, please make sure that you are not going to the endangered species or wild-caught species. Doing so will help in checking further decline of these species. Also do not purchase fur coat, purse or bag, or items made of crocodile skin or python skin. You will certainly help in preserving biodiversity by doing so.

3. Man-Wildlife Conflicts: We have discussed about the need to preserve and protect wildlife.

However, sometimes we come across conflicting situations when wildlife starts causing immense damage and danger to man and under such conditions it becomes very difficult for the forest department to pacify the affected villages and gain local support for wildlife conservation.

Instances of man animal conflicts keep on coming to lime light from several states in our country.

In Sambalpur, Orissa 195 humans were killed in the last 5 years by elephants. In retaliation the villagers killed 95 elephants in the border region of Kote-Chamarajanagar belt in Mysore have been reported recently. The man-elephant conflict in this region has arisen because of massive damage done by the elephants to the farmer's cotton and sugarcane crops. The agonized villagers electrocute the elephants and sometimes hide explosives in the sugarcane fields, which explode

as the elephants intrude into their fields. In fact, more killings are done by locals than by poachers.

ENDANGERED AND ENDEMIC SPECIES :

Endangered species A species whose numbers are reduced to the point. That means endangered species are in immediate danger of extinction.

The International Union Conservation of Nature (IUCN) classified the species of plants and animals as:

(a) Endangered species

(b) Threatened species: Species (including animals, plants, fungi, etc.) which are vulnerable to endangerment in the near future)

(c) Rare species : Among the important endangered animal species, Indian wild ass; the Kashmir stag, the Golden Langur etc .. are considered highly endangered. There are also endangered bird species like Siberian crane; the great Indian Bustard; the florican etc..

The IUCN published the data on endangered species of both plants and animals of India. The data symbolizes the working signal for those species which are endangered and if not protected are likely to become extinct in near future

A species is said to be **extinct** when it is not seen in the wild for 50 years at a stretch e.g. Dodo, Passenger Pigeon.

A species is said to be **endangered** when its number has been reduced to a critical level or whose habitat, have been drastically reduced and if such species is not protected and conserved, it is in immediate danger of extinction.

Endangered species of India:

The International Union for Conservation of Nature and Natural Resources(IUCN) publishes the **Red Data Book** which include the list of endangered species of plants and animals. The red data symbolizes the warning signal for those species which are endangered and if not protected are likely to become extinct in near future

The animals that are listed under the critically endangered category are as under:

- 1) MalabarLargeSpottedCivet
- 2) NamdaphaFlyingSquirrel
- 3) SalimAli's FruitBat
- 4) SumatranRhinoceros

EndangeredSpeciesareasunder :

- 1) AsiaticLion
- 2) AsiaticBlackBear
- 3) DesertCat
- 4) GreatIndianRhinoceros
- 5) IndianElephant(or)AsianElephant

ThreatenedSpeciesareasunder :

- 1) IndianWildAss
- 2) Leopard

Endemic species of India:

India has two biodiversity hot-spots and thus possesses a large number of endemic species. The endemic species are those taxa whose distribution is confined to a restricted area due to their specific ecological niches and edaphic gradients. Therefore, the habitats of endemic species are far more vulnerable than other species. Endemic species once lost, it is a loss of biodiversity of these species forever.

In India there are about 5725 endemic taxa of angiosperms (33.5% of Indian flora) which are located in 25 hot spots. The major hotspots in India which contain largest number of endemic plant species are the Southern Western Ghats and Eastern Himalayas with 1286 and 1808 endemic species respectively. There are about 1272 species of endemic angiosperms out of 3800 species occurring in Kerala (33.5% of Kerala flora) which represent 22.6% of Indian endemics. Seventy percent of the 1272 species of endemics have the major areas of distribution in Kerala with spill over in adjacent regions. On the basis of the study of the distributional range, about 102 endemic species occur exclusively in Kerala.

A large number out of a total of 81,000 of animals in our country is endemic. The Western Ghats are particularly rich in amphibians (frogs, toads etc) and reptiles (lizards, crocodiles etc) about 62% amphibians and 50% lizards are endemic to Western Ghats



Bicolored frog
(*Clinotarsus curtipes*)
[Ghats](#) of [India](#)

[orange-breasted sunbird](#) (*Anthobaphes violacea*)
and the [Kniphofia uvaria](#) plant is endemic to the [Western](#)

CONSERVATION OF BIODIVERSITY :

In order to maintain and conserve biodiversity, the Ministry of Environment and Forests, government of India has already taken several steps to manage wildlife, the objectives of which are:

1. Maintenance of a number of species in protected areas such as National Parks, Sanctuaries..
2. To improve the biosphere reserves
3. Implement strict restrictions of export of rare plants and animals
4. Educate the public on these through the government agencies and NGO's.

A) In-situ conservation: The preservation of species **in its natural ecosystem** is called in-situ conservation. As a consequence, protected areas are being identified and maintained for natural conservation of species by individual countries. For the conservation and management of endangered species several projects have been established.

These are:

Tiger Projects: Corbett National Park which is 300 km from New Delhi is the oldest National Park of India having 1318.54 sq km. It was one of the nine Tiger Reserves created at the launch of the Project Tiger in 1973.

Gir Lion Projects: The Gir Forest of Gujarat where lions are found. This has an area of 1412 sq kms and declared as a National Park.

Elephant Projects: The objective was to ensure long-term survival of population of elephants (not come into operations). Project Elephant (PE), a centrally sponsored scheme, was launched in February 1992 to provide financial and technical support to major elephant bearing States in the country for protection of elephants and their habitats. The Project is being implemented in 13 States/UTs, viz..Andhra Pradesh, Arunachal Pradesh, Assam, Jharkhand, Karnataka, Kerala, Meghalaya, Nagaland, Orissa, Tamil Nadu, Uttaranchal, Uttar Pradesh and West Bengal.

There are about 7000 protected areas in the world which include a variety of National parks, Sanctuaries etc which vary in size (between 100 to 500 sq km), purpose (protection of one or more species and their habitats).. In India, there are 39 National Parks and 492 wildlife sanctuaries.

National Parks: These are protected areas exclusively for wild life. Human activities like hunting, Firewood collection, timber harvesting etc... are restricted in these areas to that wild plants and animals could grow in a protected environment

The following measures should be adopted for the conservation of biodiversity:

1. Over grazing in the forest and areas of vegetation should be controlled because it may Destroy the useful rare plants.
2. The habitat of plants and animals should be conserved.
3. The natural condition of ecosystem should be studied and researched in time and again, then Specific programs for conservation should be conducted.
4. Human activities should be done without destroying natural environment.
5. Illegal hunting and smuggling of animals and plants should be strictly avoided.
6. Effective laws and rules should be adopted for the conservation of rare animals and plants.
7. Industries are established from the raw materials. During the process of collecting raw materials, care should be taken not to destroy useful plants and habitats of animals.
8. Public awareness should be created about the importance of rare animals and plants, causes of rareness and measures for their preservation.

B) Ex-situ conservation: The conservation of biodiversity out of the context of their natural habitats is referred to as ex-situ conservation

Ex-situ conservation of biodiversity involves the breeding and maintenance of endangered species in artificial ecosystems such as zoos, nurseries, botanical gardens, gene banks, etc. There is less competition for food, water and space among the organisms.

Ex-situ conservation has the following advantages:

- (a) The animals are provided with a longer time and breeding activity.
- (b) The species bred in captivity can be reintroduced in the wild.
- (c) Genetic techniques can be used for the preservation of endangered species.

Botanical gardens: Initially, botanical gardens were created for aesthetic reasons where different plants were grown for pleasure. However, over a period, the botanical gardens have become centres for scientific research for the propagation of many local and exotic plants

Zoological Parks: Zoos are places where animals are kept in captivity. Many such animals are rare or endangered and in many cases, extinct in their natural habitat. The Zoos play an important role in the breeding of rare and endangered species. Sometimes, after successful breeding, many species are released in their natural environment. Thus, the Zoos are a major contributor in checking the extinction of the endangered species. The zoos also significantly contribute in creating awareness and interest in wildlife, specially among the children.

Gene banks/ Seed banks: These are cold storages where seeds are kept under controlled temperature and humidity for storage. This is an important way of preserving the genetic resources. Seeds preserved under controlled conditions (minus temperature), remain viable for long durations of time. Besides storage, gene banks are also well equipped for scientific researches related to seed processing, seed drying and testing.

In India we have the following important gene and seed bank facilities.

i) National Bureau of Plant Genetic Resources (NBPGR) is located in New Delhi. Here agricultural and horticultural crops are stored by **cryopreservation** of seeds, pollens etc.

Cryopreservation: This is the newest application of technology for preservation of biotic parts. Here, the organism or their parts are stored in liquid nitrogen (-196 degree C).

ii) National Bureau of Animal Genetic Resources (NBAGR) located at Karnal, Haryana. It preserves the semen of domesticated bovine animals.

NATIONAL BIODIVERSITY ACT :

The Biological Diversity Act, 2002, is an act of the Parliament of India enacted to conserve biological diversity, ensure its sustainable use, and promote fair and equitable benefit-sharing from the utilization of biological resources and associated knowledge. It establishes a three-tiered structure for its implementation: the National Biodiversity Authority (NBA) at the national level, State Biodiversity Boards (SBBs) at the state level, and Biodiversity Management Committees (BMCs) at the local level.

Features and Objectives:

Conservation:

The Act aims to conserve biological diversity within India, including its components and associated ecosystems.

Sustainable Use:

It promotes the sustainable use of biological resources, ensuring their availability for present and future generations.

Benefit Sharing:

The Act mandates the fair and equitable sharing of benefits derived from the use of biological resources and traditional knowledge with local communities and indigenous people who have traditionally conserved and used these resources.

Regulation of Access:

It regulates access to biological resources and associated knowledge, requiring approval from the NBA for access by foreign individuals, institutions, or companies.

Intellectual Property Rights:

The Act requires approval from the NBA before applying for any intellectual property rights (IPR) based on research or information on biological resources obtained from India.

Benefit Claimers:

The Act identifies benefit claimers as conservers of biodiversity, or creators or holders of associated traditional knowledge, and ensures their participation in benefit-sharing mechanisms.

Offences and Penalties:

The Act outlines offences for violations, such as failing to obtain necessary approvals or providing prior intimation for certain activities. Violations can lead to imprisonment and/or fines.

The Biological Diversity Act, 2002, is a crucial piece of legislation for India's efforts to conserve its rich biodiversity and ensure its sustainable use for the benefit of present and future generations.

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